

№₀ Weekly Problem

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Problem

Prove that there is no polynomial $f(x)$ with integer coefficients, so that $f(7) = 11$ and $f(11) = 13$.

Solution

First, we prove the following lemma that will be helpful later.

Lemma. *Let $f(x)$ be a polynomial with integer coefficients, and let s and t be integers. Then, $f(s) - f(t)$ is divisible by $s - t$.*

Proof. Let $f(x) = a_n x^n + a_{n-1} x^{n-1} + a_{n-2} x^{n-2} + \cdots + a_0 x^0$ where $a_i \in \mathbb{Z}$ for all $i \in \{0, \dots, n\}$. Then,

$$f(s) - f(t) = a_n(s^n - t^n) + a_{n-1}(s^{n-1} - t^{n-1}) + \cdots + a_1(s - t).$$

Notice that $(s^k - t^k) = (s - t)(s^{k-1}t^0 + s^{k-2}t^1 + s^{k-3}t^2 + \cdots + s^0t^{k-1})$ for all positive integers k . Therefore, $f(s) - f(t)$ is divisible by $s - t$. □

Now we turn our attention to proving that there is no polynomial $f(x)$ with integer coefficients, so that $f(7) = 11$ and $f(11) = 13$.

Proof. Suppose, for sake of finding a contradiction, that there did exist a polynomial $f(x)$ having all integer coefficients such that $f(7) = 11$ and $f(11) = 13$. But this immediately contradicts the lemma above, as $f(7) - f(11) = -2$ is not divisible by $7 - 11 = -4$. □